

Inst: Represent the given image's region for classification



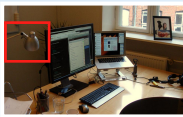
MSCOCO
Box
classification

Only map the region to
simple class name

C_{t1} : potted plant

C_{t2} : toilet

Inst: Fine the best image caption for the
following image's region



FG-OVD
Visual
Understanding

The text candidate is
still short and simple

C_{t1} : A lamp made of metal with a white pipe, grey finial, and a grey shade.

(a) Existing Benchmarks

Inst: Fine an image caption
describing the following
image's region.



task2, $i \rightarrow t$

Detailed caption
for a region

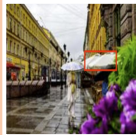
C_t : A marble sculpture of a standing figure draped in flowing robes, holding a book in the left hand and a staff in the right hand...

Inst: Show me an image's region that best
captures the following common scene
description.

C_t : A large, white, rectangular umbrella with a slightly curved top and a taut fabric surface. The umbrella has a visible seam running along the edge...

task1, $t \rightarrow i$

Find the region by
specific description



Inst: Retrieve a image that aligns with the
modification instructions and has the same
background



q_t : Same white
flowers in a metal
vase on a table.

task4, $it \rightarrow i$

Image editing pairs
requiring context cues



Inst: Retrieve the similar product as the
one bounded by the red box.



task3, $i \rightarrow i$

The same product
under different light



(b) FGMB Benchmark